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SEMICONDUCTOR DEVICE AND METHOD OF FORMING THE SAME

Abstract

The present disclosure provides a semiconductor device. The semiconductor device includes a substrate, an electronic fuse, and a switching transistor. The electronic fuse is on the substrate. The switching transistor is on the substrate and next to the electronic fuse, in which a first doping region of the substrate underneath the electronic fuse has a first conductive type, a second doping region of the substrate underneath the switching transistor has a second conductive type, the first conductive type and the second conductive type are the same, the first doping region of the substrate is disposed between third doping regions of the substrate, and third conductive types of the third doping regions are different from the first conductive type.

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Background/Summary

BACKGROUND

Field of Invention

[0001] The present disclosure relates to a semiconductor device and a method of forming the same.

Description of Related Art

[0002] Electronic fuses are designed as sacrificial components and may be used in semiconductor devices to protect the semiconductor devices from being damaged and/or improve the performance efficiency of the semiconductor devices. For example, the electronic fuse is designed to be blown when too much electric flow is passing through the electronic fuse, thereby protecting the semiconductor from being damaged by a high voltage. In addition, by blowing the electronic fuse(s) in the semiconductor device, a flow path of the current in the semiconductor may change to provide various circuits depending on the requirements. In some situations, when a component fails to work in the semiconductor device, the electronic fuse connected to that component may be blown to avoid slowing down the performance efficiency of the semiconductor devices. However, as the dimension of the semiconductor device shrinks, distances between the electronic fuse and other components also become smaller. When the electronic fuse is packed too close to the nearby component and the types of dopants implanted under the electronic fuse and the nearby component are different, the dopants may diffuse and cause contamination, thereby perishing the function of the electronic fuse and/or the nearby component. Therefore, it is necessary to develop a new semiconductor device including the electronic fuse and a method of forming the same to satisfy every aspect.

SUMMARY

[0003] The present disclosure provides a semiconductor device. The semiconductor device includes a substrate, an electronic fuse, and a switching transistor. The electronic fuse is on the substrate. The switching transistor is on the substrate and next to the electronic fuse, in which a first doping region of the substrate underneath the electronic fuse has a first conductive type, a second doping region of the substrate underneath the switching transistor has a second conductive type, the first conductive type and the second conductive type are the same, the first doping region of the substrate is disposed between third doping regions of the substrate, and third conductive types of the third doping regions are different from the first conductive type.

[0004] In some embodiments, the first doping region and the third doping regions are disposed between source/drain doping regions of the substrate.

[0005] In some embodiments, depths of the source/drain doping regions in the substrate are larger than depths of the third doping regions in the substrate.

[0006] In some embodiments, dopant concentrations of the source/drain doping regions are larger than dopant concentrations of the third doping regions.

[0007] In some embodiments, fourth conductive types of the source/drain doping regions are the same as the third conductive types.

[0008] In some embodiments, the second doping region of the substrate is disposed between fourth doping regions of the substrate, fourth conductive types of the fourth doping regions are different from the second conductive type, and one of the source/drain doping regions extends continuously from one of the third doping regions to one of the fourth doping regions.

[0009] In some embodiments, the first doping region is in direct contact with the third doping regions.

[0010] In some embodiments, the electronic fuse includes a gate structure, first spacers on sidewalls of the gate structure, and second spacers on the first spacers.

[0011] In some embodiments, the first doping region is disposed below the gate structure and the first spacers, and the third doping regions are disposed below the second spacers.

[0012] In some embodiments, the first conductive type is a P type, the second conductive type is a P type, and the third conductive types are N types; or the first conductive type is an N type, the

second conductive type is an N type, and the third conductive types are P types.
[0013] In some embodiments, a minimum distance between a gate structure of the electronic fuse and a gate structure of the switching transistor is smaller than 100 nm.
[0014] The present disclosure also provides a method of forming semiconductor device. The method includes the following operations. A first ion implantation process is performed to form a first implantation region in the substrate. A gate structure of an electronic fuse and a gate structure of a switching transistor are formed on the first implantation region. First spacers are formed on sidewalls of the gate structure of the electronic fuse. A second ion implantation process is performed adjacent to the first spacers to transform first portions of the first implantation region into second implantation regions in the substrate, in which conductive types of the second implantation regions are different from a conductive type of the first implantation region. Second spacers are formed on the first spacers. A third ion implantation process is performed adjacent to the second spacers to transform first portions of the second implantation regions into third implantation regions in the substrate, in which conductive types of the third implantation regions are the same as conductive types of the second implantation regions.
[0015] In some embodiments, after performing the third ion implantation process, a second portion of the first implantation region is remained below the gate structure and the first spacers, second portions of the second implantation regions are remained below the second spacers, and the second portion of the first implantation region and the second portions of the second implantation regions are located between the third implantation regions.
[0016] In some embodiments, an ion implantation energy used in the third ion implantation process is larger than an ion implantation energy used in the second ion implantation process.
[0017] In some embodiments, a dopant concentration used in the second ion implantation process is from 1×10.sup.13 atoms/cm.sup.3 to 1×10.sup.15 atoms/cm.sup.3, and a dopant concentration used in the third ion implantation process is from 1×10.sup.14 atoms/cm.sup.3 to 1×10.sup.16 atoms/cm.sup.3.
[0018] In some embodiments, forming the first spacers further includes forming the first spacers on sidewalls of the gate structure of the switching transistor, and after performing the third ion implantation process, second portions of the second implantation regions are remained and located adjacent to the gate structure of the electronic fuse, third portions of the second implantation regions are remained and located adjacent to the gate structure of the switching transistor, and one of the third implantation regions extends continuously from one of the second portions of the second implantation regions to one of the third portions of the second implantation regions.
[0019] In some embodiments, a P type dopant is used in the first ion implantation process, and N type dopants are used in the second ion implantation process and the third ion implantation process; or an N type dopant is used in the first ion implantation process, and P type dopants are used in the second ion implantation process and the third ion implantation process.
[0020] In some embodiments, a dopant concentration used in the first ion implantation process is from 1×10.sup.11 atoms/cm.sup.3 to 1×10.sup.14 atoms/cm.sup.3.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0021] The present disclosure can be more fully understood by reading the following detailed description of the embodiments, with reference made to the accompanying figures as follows.
[0022] FIG. **1** is a cross-sectional view diagram of a semiconductor device of the present disclosure according to some embodiments.
[0023] FIG. **2** is a top-view diagram of a semiconductor device of the present disclosure according to some embodiments.

[0024] FIG. **3** is a flow chart of a method of forming a semiconductor device of the present disclosure according to some embodiments.
[0025] FIGS. **4** to **10** are cross-sectional view diagrams of the structures during the formation of a semiconductor device using a method of the present disclosure according to some embodiments.

DETAILED DESCRIPTION

[0026] To make the description of the present disclosure detailed and complete, the following is an illustrative description of the aspects of the embodiments. This is not to limit the embodiments of the present disclosure to only one form. The embodiments of the present disclosure may be combined or substituted with each other when it is beneficial, and other embodiments may be added without further explanation.
[0027] In addition, spatially relative terms, such as below and above, etc., may be used in the present disclosure to describe the relationship between one element (or feature) to another element (or feature) in the figures. In addition to the orientation depicted in the figures, spatially relative terms are intended to encompass different orientations of the device in use or in operation. For example, the device may be oriented otherwise (e.g., rotated at 90 degrees), and the spatially relative terms can be interpreted accordingly. In the present disclosure, unless otherwise indicated, the same element numbers in different figures refer to the same or similar elements formed from the same or similar materials by the same or similar methods.
[0028] The terms “around”, “approximately”, “nearly”, “basically”, “substantially”, etc., used in the present disclosure include the stated values (or characteristics) and a deviation of the stated values (or characteristics) understood by one skilled in the art. For example, considering the errors of the values (or characteristics), these terms may indicate the values within one or more standard deviations (e.g., the values within ±30%, ±20%, ±15%, ±10%, or ±5%), or may indicate the characteristics including the deviation from the practical operation (e.g., the “substantially parallel” may indicate close to parallel in practical, rather than a perfect ideally parallelism). Furthermore, it is possible to select an acceptable range of the deviation according to the nature of the measurement or other properties, instead of applying only one single deviation range to all the values (or characteristics).
[0029] The present disclosure provides a semiconductor device **10**, as shown in FIGS. **1** and **2**, in which FIG. **1** is a cross-sectional view of FIG. **2** along a line A-A. The semiconductor device **10** includes a substrate **101**, an electronic fuse **102**, and switching transistors **103**. The electronic fuse **102** is disposed on the substrate **101**. The switching transistors **103** are disposed on the substrate **101** and next to the electronic fuse **102**, in which a first doping region D**1** of the substrate **101** underneath the electronic fuse **102** has a first conductive type, a second doping region D**2** of the substrate **101** underneath each of the switching transistors **103** has a second conductive type, the first conductive type and the second conductive type are the same, the first doping region D**1** of the substrate **101** is disposed between third doping regions D**3** of the substrate **101**, and third conductive types of the third doping regions D**3** are different from the first conductive type. Since the first conductive type of the first doping region D**1** and the second conductive type of the second doping region D**2** disposed underneath the electronic fuse **102** and the switching transistor **103** are the same, the dopant of the first doping region D**1** and the dopant of the second doping region D**2** may not be contaminated with each other during the formation of the semiconductor device **10**, considering the dopant may diffuse between regions, especially when the distance (e.g., a minimum distance W**1** described below) between the electronic fuse **102** and the switching transistor **103** is small (e.g., smaller than 100 nm). The formation of the semiconductor device **10** may also be simplified by forming the first doping region D**1** and the second doping region D**2** together in a single implantation process (e.g., a first ion implantation process **201** described below). Moreover, when the distance between the electronic fuse **102** and the switching transistor **103** becomes smaller, compared with separately implanting a small area of the first doping region D**1** and a small area of the second doping region D**2**, the single implantation process is much more favorable by

implanting a large area (e.g., a dashed line **108** indicated by the embodiment shown in FIG. **2**) of the first doping region D**1** and the second doping region D**2** in the process, considering the manufacturing tool nowadays may not be able to form a small enough area of the doping region. The method **20** of the present disclosure is discussed in detail in the following. In addition, the third doping regions D**3** beside the first doping region D**1** can enable the current to flow more easily in the first doping region D**1** when applying a voltage to a gate structure **102**G of the electronic fuse **102**. Next, the semiconductor device **10** of the present disclosure is described in detail with the following embodiments.

[0030] The substrate **101** may be any suitable substrate. In some embodiments, the substrate **101** may be a semiconductor substrate and may include a semiconductor material including an elemental semiconductor material, for example, carbon, monocrystalline silicon, polycrystalline silicon, amorphous silicon, germanium, tin, sulfur, selenium, tellurium, or the like; a compound semiconductor material, for example, silicon carbide, nitride boron, aluminum nitride, gallium nitride, gallium phosphide, gallium arsenide, indium phosphide, indium arsenide, indium antimonide, zinc oxide, or the like; an alloy semiconductor material, for example, SiGe, AlGaAs, InGaAs, InGaP, AlInAs, GaAsP, AlGaN, InGaN, AlGaInP, or the like; or combinations thereof.

[0031] The substrate **101** includes different doping regions. For example, the first doping region D**1** is underneath the electronic fuse **102** to provide a channel for the current to flow underneath the electronic fuse **102**. For example, the second doping region D**2** is underneath the switching transistor **103** to provide a channel for the current to flow underneath the switching transistor **103**. For example, the third doping regions D**3** beside the first doping region D**1** enables the current to flow more easily in the first doping region D**1** when applying a voltage to the gate structure **102**G of the electronic fuse **102**. In some embodiments, the substrate **101** further includes fourth doping regions D**4**, and the second doping region D**2** is disposed between the fourth doping regions D**4**. In some embodiments, the first doping region D**1** and the third doping regions D**3** are disposed between source/drain doping regions D**5**, and the second doping region D**2** and the fourth doping regions D**4** are also disposed between source/drain doping regions D**5**.

[0032] In some embodiments, the first doping region D**1** is disposed underneath the gate structure **102**G of the electronic fuse **102** and underneath the first spacers S**1** disposed on the gate structure **102**G of the electronic fuse **102**. In some embodiments, the second doping region D**2** is disposed underneath the gate structure **103**G of the switching transistor **103** and underneath the first spacers S**1** disposed on the gate structure **103**G of the switching transistor **103**. In some embodiments, the third doping regions D**3** are disposed underneath the second spacers S**2** disposed on the first spacers S**1** that are disposed on the gate structure **102**G of the electronic fuse **102**. In some embodiments, the fourth doping regions D**4** are disposed underneath the second spacers S**2** disposed on the first spacers S**1** that are disposed on the gate structure **103**G of the switching transistor **103**. In some embodiments, in a portion of the substrate **101** that is disposed underneath a space or a region between the gate structure **102**G of the electronic fuse **102** and the gate structure **103**G of the switching transistor **103**, one of the source/drain doping regions D**5** extends continuously from one of the third doping regions D**3** to one of the fourth doping regions D**4**, such that the one of the source/drain doping regions D**5** may be shared by the electronic fuse **102** and the switching transistor **103** nearby. In some embodiments, the first doping region D**1** is in direct contact with the third doping regions D**3**. In some embodiments, the second doping region D**2** is in direct contact with the fourth doping regions D**4**. In some embodiments, depths D**5**′ (e.g., along a third direction Z that is perpendicular to a first direction X and a second direction Y) of the source/drain doping regions D**5** in the substrate **101** are larger than depths D**3**′ (e.g., along the third direction Z) of the third doping regions D**3** in the substrate **101**. In some embodiments, a ratio of the depth D**5**′ of either one of the source/drain doping regions D**5** to the depth D**3**′ of either one of the third doping regions D**3** is preferably from 1.5 to 7.5, for example, 1.5, 2.5, 3.5, 4.5, 5.5, 6.5, or 7.5, to have enough doping amounts of the source/drain doping regions D**5** and the third doping

regions D**3**, have a clear boundary between the working regions of and the source/drain doping regions D**5** and the third doping regions D**3**, and avoid wasting too many manufacturing resources while not having a significant improvement.
[0033] The first doping region D**1** has the first conductive type, the second doping region D**2** has the second conductive type, the third doping regions D**3** have the third conductive types, the fourth doping regions D**4** have the fourth conductive types, and the source/drain doping regions D**5** have the fifth conductive types. The first conductive type and the second conductive type are the same. The third conductive types, the fourth conductive types, and the fifth conductive types are than same and different from the first conductive type and the second conductive type. In some embodiments, the first doping region D**1** and the second doping region D**2** may respectively include a P type dopant including boron, gallium, or the like, such that the first conductive type and the second conductive type are P types, and the third doping regions D**3**, the fourth doping regions D**4**, and the source/drain doping regions D**5** may respectively include an N type dopant including phosphorus, antimony, arsenic, or the like, such that the third conductive types, the fourth conductive types, and the fifth conductive types are N types. In some embodiments, the first doping region D**1** and the second doping region D**2** may respectively include an N type dopant including phosphorus, antimony, arsenic, or the like, such that the first conductive type and the second conductive type are N types, and the third doping regions D**3**, the fourth doping regions D**4**, and the source/drain doping regions D**5** may respectively include a P type dopant including boron, gallium, or the like, such that the third conductive types, the fourth conductive types, and the fifth conductive types are P types. In some embodiments, a dopant concentration of either one of the first doping region D**1** and the second doping region D**2** is preferably from 1×10.sup.11 atoms/cm.sup.3 to 1×10.sup.14 atoms/cm.sup.3, for example, 1×10.sup.11 atoms/cm.sup.3, 1×10.sup.12 atoms/cm.sup.3, 1×10.sup.13 atoms/cm.sup.3, or 1×10.sup.14 atoms/cm.sup.3. In some embodiments, dopant concentrations of the source/drain doping regions D**5** are larger than dopant concentrations of the third doping regions D**3** and dopant concentrations of the fourth doping regions D**4**. In some embodiments, a dopant concentration of either one of the third doping regions D**3** and either one of the fourth doping regions D**4** is preferably from 1×10.sup.13 atoms/cm.sup.3 to 1×10.sup.15 atoms/cm.sup.3, for example, 1×10.sup.13 atoms/cm.sup.3, 5×10.sup.13 atoms/cm.sup.3, 1×10.sup.14 atoms/cm.sup.3, 5×10.sup.14 atoms/cm.sup.3, or 1×10.sup.15 atoms/cm.sup.3. In some embodiments, a dopant concentration of either one of the source/drain doping regions D**5** is preferably from 1×10.sup.14 atoms/cm.sup.3 to 1×10.sup.16 atoms/cm.sup.3, for example, 1×10.sup.14 atoms/cm.sup.3, 5×10.sup.14 atoms/cm.sup.3, 1×10.sup.15 atoms/cm.sup.3, 5×10.sup.15 atoms/cm.sup.3, or 1×10.sup.16 atoms/cm.sup.3.
[0034] In some embodiments, since the first conductive type of the first doping region D**1** and the second conductive type of the second doping region D**2** disposed underneath the electronic fuse **102** and the switching transistor **103** are the same, a minimum distance W**1** between the gate structure **102**G of the electronic fuse **102** and the gate structure **103**G of the switching transistor **103** may be reduced without making the dopants in the first doping region D**1** and the second doping region D**2** contaminant with each other. In some embodiments, the minimum distance W**1** is preferably in a range between 10 nm and 100 nm, for example, 20 nm, 30 nm, 40 nm, 50 nm, 60 nm, 70 nm, 80 nm, or 90 nm, to effectively reduce the size of the semiconductor device **10** and to avoid a significant interference between the gate structure **102**G of the electronic fuse **102** and the gate structure **103**G of the switching transistor **103**.
[0035] In some embodiments, the gate structure **102**G of the electronic fuse **102** is disposed between a source structure (not shown) and a drain structure (not shown), and the source structure and the drain structure are disposed on the source/drain doping regions D**5** adjacent to the electronic fuse **102**. In some embodiments, the gate structure **103**G of the switching transistor **103** is disposed between a source structure (not shown) and a drain structure (not shown), and the source structure and the drain structure are disposed on the source/drain doping regions D**5**

adjacent to the switching transistor **103**.

[0036] In some embodiments, an insulating layer **104** is disposed on the substrate **101**, between the substrate **101** and the gate structure **102**G of the electronic fuse **102**, and between the substrate **101** and the gate structure **103**G of the switching transistor **103**. In some embodiments, the insulating layer **104** includes any suitable material, for example, silicon oxide.

[0037] In some embodiments, a first electrode **105** is disposed on the source/drain doping regions D**5** and on the side of the switching transistor **103** opposite the side facing the electronic fuse **102**. The first electrode **105** can be the electrical ground during blowing out the electronic fuse **102** and be used to measure the electrical resistance when determining whether the electronic fuse **102** is blown out or not. In some embodiments, the first electrode **105** includes any suitable material, for example, a metal.

[0038] The electronic fuse **102** may be blown out when a high voltage is applied to the gate structure **102**G of the electronic fuse **102**. For example, when the high voltage is applied to the gate structure **102**G of the electronic fuse **102**, a portion P**1** and a portion P**2** of the insulating layer **104** next to the gate structure **102**G of the electronic fuse **102** may respectively have probabilities to be blown out. When the portion P**1**, the portion P**2**, or a combination thereof is blown out, an electrical short happens underneath the electronic fuse **102**. To confirm whether the electrical short happens underneath the electronic fuse **102** or not, a voltage smaller than the voltage to blow out the electronic fuse **102** and larger than a minimum voltage to overcome the p-n junction underneath the electronic fuse **102** can be applied to the gate structure **102**G of the electronic fuse **102**, and an electrical resistance can be measured at the first electrode **105**. When the measured electrical resistance is small enough, the electrical short is determined to happen underneath the electronic fuse **102**.

[0039] The switching transistor **103** is configured to open or close the channel underneath the switching transistor **103**, thereby controlling whether to perform a blown-out operation to the electronic fuse **102** and whether to measure the electrical resistance underneath the electronic fuse **102**. For example, when the channel underneath the switching transistor **103** is opened in the blown-out operation, the current of the high voltage can flow from the channel underneath the electronic fuse **102** to the channel underneath the switching transistor **103** and to the electrical ground, such that the electronic fuse **102** has the probability to be blown out. However, when the channel underneath the switching transistor **103** is closed in the blown-out operation, the current of the high voltage is blocked to flow to the electrical ground, such that the electronic fuse **102** fails to be blown out. For example, when the channel underneath the switching transistor **103** is opened during the measurement of the electrical resistance, the current underneath the electronic fuse **102** can flow to the first electrode **105** and be measured. However, when the channel underneath the switching transistor **103** is closed during the measurement of the electrical resistance, the current underneath the electronic fuse **102** fails to flow to the first electrode **105** and be measured.

[0040] In some embodiments, the semiconductor device **10** may include a unit **100** including one electronic fuse **102**, two switching transistors **103** (e.g., the switching transistor **103**A and the switching transistor **103**B), and two first electrodes **105** (e.g., the first electrode **105**A and the first electrode **105**B), as shown in FIG. **1**. Therefore, whether the portion P**1** of the insulating layer **104** on the first side (e.g., the left-hand side in FIG. **1**) of the electronic fuse **102** is blown out or not may be determined by the switching transistor **103**A and the first electrode **105**A on the same first side of the electronic fuse **102**, whether the portion P**2** of the insulating layer **104** on the second side (e.g., the right-hand side in FIG. **1**) of the electronic fuse **102** is blown out or not may be determined by the switching transistor **103**B and the first electrode **105**B on the same second side of the electronic fuse **102**, and the first side and the second side are substantially opposite to each other. Since the two switching transistors **103** may work independently by opening or closing each corresponding channel, whether the portion P**1** and the portion P**2** are blown out or not can be measured independently.

[0041] The number of the units **100** is not limited. In some embodiments, the number of the units **100** is preferably an integral in a range from 1 to 20, for example, 1, 4, 8, 12, 16, or 20 (e.g., 16 shown in FIG. **2**). In some embodiments, the units **100** are arranged along the first direction X, and the electronic fuse **102**, the switching transistors **103**, and the first electrodes **105** in each of the units **100** are arranged along the second direction Y perpendicular to the first direction X. In some embodiments, the electronic fuse **102**, the switching transistors **103**, and the first electrodes **105** in one of the units **100** are respectively aligned with the electronic fuse **102**, the switching transistors **103**, and the first electrodes **105** in another one of the units **100**. In some embodiments, the switching transistor **103**A and the switching transistor **103**A in the same one of the units **100** are physically connected from the top view, as shown in FIG. **2**. In some embodiments, the electronic fuses **102** and the switching transistors **103** in different units **100** are physically connected from the top view, as shown in FIG. **2**. In some embodiments, second electrodes **106** may be common electrodes for the electronic fuses **102** that are physically connected, such that a voltage to blow out the electronic fuses **102** and/or a voltage to read the electrical resistance of the electronic fuses **102** may be simultaneously applied to the electronic fuses **102**. In some embodiments, the second electrodes **106** disposed on the electronic fuses **102** are arranged along the first direction X and the electronic fuses **102** that are physically connected are disposed between the second electrodes **106**. In some embodiments, third electrodes **107** may be common electrodes for the switching transistors **103** that are physically connected, such that the switching transistors **103** in different units **100** may be simultaneously opened and/or closed. In some embodiments, the third electrodes **107** disposed on the switching transistors **103** include third electrodes **107**A and third electrodes **107**B, in which the switching transistors **103**A are disposed between the third electrodes **107**A, and the switching transistors **103**B are disposed between the third electrodes **107**B. It is noted that since each of the units **100** includes the first electrode **105**A and the first electrode **105**B, the units **100** may work independently, for example, by selectively electrically grounding some first electrodes **105** and/or selectively reading the electrical resistance of some first electrodes **105**.

[0042] In some embodiments, each of the units **100** is disposed on a corresponding one of the active regions **101**A of the substrate **101**, and different active regions **101**A are separated from each other by an isolation region of the substrate **101**. In some embodiments, in each of the units, a width W**2** of the active region **101**A underneath the electronic fuse **102** is smaller than a width W**3** of the active region **101**A underneath the switching transistor **103**, such that the channel underneath the switching transistor **103** is larger than the channel underneath the electronic fuse **102**, and the electronic fuse **102** may be more easily blown out, considering a smaller channel underneath the electronic fuse **102**.

[0043] The present disclosure also provides a method **20** of forming the semiconductor device **10** described above. The method **20** shown in the flow chart of FIG. **3** includes an operation **21** to an operation **26**. The operation **21** includes performing a first ion implantation process **201** to form a first implantation region **202** in the substrate **101**. The operation **22** includes forming the gate structure **102**G of the electronic fuse **102** and the gate structure **103**G of the switching transistor **103** on the first implantation region **202**. The operation **23** includes forming the first spacers S**1** on sidewalls of the gate structure **102**G of the electronic fuse **102**. The operation **24** includes performing a second ion implantation process **206** adjacent to the first spacers S**1** to transform first portions **202**A of the first implantation region **202** into second implantation regions **207** in the substrate **101**, in which conductive types of the second implantation regions **207** are different from a conductive type of the first implantation region **202**. The operation **25** includes forming the second spacers S**2** on the first spacers S**1**. The operation **26** includes performing a third ion implantation process **209** adjacent to the second spacers S**2** to transform first portions **207**A of the second implantation regions **207** into third implantation regions (i.e., the source/drain doping regions D**5** described above) in the substrate **101**, in which conductive types of the third implantation regions are the same as conductive types of the second implantation regions **207**.

Next, the method **20** of the present disclosure is described in detail with the following embodiments.

[0044] The operation **21** includes performing the first ion implantation process **201** to form the first implantation region **202** in the substrate **101**, as shown in FIG. **4**. Portions (e.g., the first potions **202**A in FIG. **7**) of the first implantation region **202** will be transformed to the second implantation regions **207**, and portions of the first implantation region **202** remained underneath the gate structure **102**G of the electronic fuse **102**, the gate structure **103**G of the switching transistor **103**, and the first spacers S**1** are the first doping region D**1** and the second doping region D**2** described above. Since the first doping region D**1** and the second doping region D**2** are implanted together in the first ion implantation process **201**, the implantation area (e.g., a dashed line **108** indicated by the embodiment shown in FIG. **2**) performed by the first ion implantation process **201** can be larger than the implantation areas performed by separately implanting the first doping region D**1** and the second doping region D**2**. Therefore, the manufacturing tools of forming the separated implantation areas of the first doping region D**1** and the second doping region D**2** are not required in the present disclosure to avoid reaching the limitation of the manufacturing tools. For example, the pattern formed by the lithography tool may reach a size limitation and cannot be smaller, such that the size of the implantation area based on the pattern formed by the lithography tool is limited. In addition, since the first doping region D**1** and the second doping region D**2** are implanted together in the first ion implantation process **201**, the conductive type of the first doping region D**1** and the conductive type of the second doping region D**2** are the same. In some embodiments, the P type dopant (e.g., boron, gallium, or the like) is used in the first ion implantation process **201**. In some embodiments, the N type dopant (e.g., phosphorus, antimony, arsenic, or the like) is used in the first ion implantation process **201**. In some embodiments, a dopant concentration used in the first ion implantation process **201** is preferably from 1×10.sup.11 atoms/cm.sup.3 to 1×10.sup.14 atoms/cm.sup.3, for example, 1×10.sup.11 atoms/cm.sup.3, 1×10.sup.12 atoms/cm.sup.3, 1×10.sup.13 atoms/cm.sup.3, or 1×10.sup.14 atoms/cm.sup.3. In some embodiments, as shown in FIG. **5**, after forming the first implantation region **202**, the insulating layer **104** is formed on the first implantation region **202** by any suitable deposition method, for example, a chemical deposition process or a physical deposition process.

[0045] The operation **22** includes forming the gate structure **102**G of the electronic fuse **102** and the gate structure **103**G of the switching transistor **103** on the first implantation region **202**, as shown in FIGS. **5** to **6**. In some embodiments, a material **203** of the gate structure **102**G and the gate structure **103**G is formed on the first implantation region **202** by any suitable deposition method, for example, a chemical deposition process or a physical deposition process. Next, a patterned photoresist layer **204** is formed on the material **203** to define the pattern of the gate structure **102**G and the gate structure **103**G. Specifically, portions of the material **203** exposed by openings **2040** of the patterned photoresist layer **204** are removed by any suitable etching method, for example, a dry etching process or a wet etching process, to form the gate structure **102**G and the gate structure **103**G. In some embodiments, after forming the gate structure **102**G and the gate structure **103**G, the photoresist layer **204** is removed.

[0046] The operation **23** includes forming the first spacers S**1** on sidewalls of the gate structure **102**G of the electronic fuse **102**, as shown in FIGS. **7** to **8**. In some embodiments, forming the first spacers S**1** further includes forming the first spacers S**1** on sidewalls of the gate structure **103**G of the switching transistor **103**. Specifically, a material **205** of the first spacers S**1** is conformally formed on the gate structure **102**G and the gate structure **103**G by any suitable deposition method, for example, an atomic layer deposition process. Next, portions of the material **205** are removed by any suitable etching method, for example, a dry etching process or a wet etching process, to remain portions of the material **205** on the sidewalls of the gate structure **102**G and the sidewalls of the gate structure **103**G. In some embodiments, the material **205** may be any suitable material, for example, silicon nitride. The first spacers S**1** will be used to define the position of the second implantation regions **207** in the following operation and portions of the substrate **101** underneath the first spacers S**1** can be used to provide tolerance for the diffusion of the second implantation regions **207**.

[0047] The operation **24** includes performing the second ion implantation process **206** adjacent to the first spacers S**1** to transform the first portions **202**A of the first implantation region **202** into the second implantation regions **207** in the substrate **101**, as shown in FIGS. **7** to **8**, in which conductive types of the second implantation regions **207** are different from a conductive type of the first implantation region **202**. Portions (e.g., the first potions **207**A in FIG. **9**) of the second implantation regions **207** will be transformed to the third implantation regions (i.e., the source/drain doping regions D**5**), and portions of the second implantation regions **207** remained underneath the second spacers S**2** are the third doping regions D**3** and the fourth doping regions D**4** described above. In the embodiments that the P type dopant is implanted in the first implantation region **202**, the N type dopant (e.g., phosphorus, antimony, arsenic, or the like) is used in the second ion implantation process **206**. In the embodiments that the N type dopant is implanted in the first implantation region **202**, the P type dopant (e.g., boron, gallium, or the like) is used in the second ion implantation process **206**. In some embodiments, a dopant concentration used in the second ion implantation process **206** is preferably from 1×10.sup.13 atoms/cm.sup.3 to 1×10.sup.15 atoms/cm.sup.3, for example, 1×10.sup.13 atoms/cm.sup.3, 5×10.sup.13 atoms/cm.sup.3, 1×10.sup.14 atoms/cm.sup.3, 5×10.sup.14 atoms/cm.sup.3, or 1×10.sup.15 atoms/cm.sup.3. In some embodiments, an ion implantation energy used in the second ion implantation process **206** is preferably from 2 KeV to 10 KeV, for example, 2 KeV, 4 KeV, 6 KeV, 8 KeV, or 10 KeV, such that a desirable depth (e.g., the depths D**3**′ described above) of the second implantation regions **207** can be achieved.

[0048] The operation **25** includes forming the second spacers S**2** on the first spacers S**1**, as shown in FIGS. **9** to **10**. Specifically, a material **208** of the second spacers S**2** is conformally formed on the first spacers S**1**, the gate structure **102**G and the gate structure **103**G by any suitable deposition method, for example, an atomic layer deposition process. Next, portions of the material **208** are removed by any suitable etching method, for example, a dry etching process or a wet etching process, to remain portions of the material **208** on the first spacers S**1**. In some embodiments, the material **208** may be any suitable material, for example, silicon nitride. The second spacers S**2** will be used to define the position of the third implantation regions (i.e., the source/drain doping regions D**5**) in the following operation and portions of the substrate **101** underneath the second spacers S**2** can be used to provide tolerance for the diffusion of the third implantation regions.

[0049] The operation **26** includes performing the third ion implantation process **209** adjacent to the second spacers S**2** to transform the first portions **207**A of the second implantation regions **207** into the third implantation regions (i.e., the source/drain doping regions D**5**) in the substrate **101**, as shown in FIG. **10**, in which conductive types of the third implantation regions are the same as conductive types of the second implantation regions **207**. After performing the third ion implantation process **209**, portions of the first implantation region **202** remained underneath the gate structure **102**G and the gate structure **103**G are respectively the first doping regions D**1** and the second doping regions D**2** described above, portions of the second implantation regions **207** remained underneath the second spacers S**2** are the third doping regions D**3** and the fourth doping regions D**4** described above, and the third implantation regions are the source/drain doping regions D**5** described above. In the embodiments that the P type dopant is implanted in the first implantation region **202**, the N type dopants (e.g., phosphorus, antimony, arsenic, or the like) are used in the second ion implantation process **206** and the third ion implantation process **209**. In the embodiments that the N type dopant is implanted in the first implantation region **202**, the P type dopants (e.g., boron, gallium, or the like) are used in the second ion implantation process **206** and the third ion implantation process **209**. In some embodiments, a dopant concentration used in the third ion implantation process **209** is preferably from 1×10.sup.14 atoms/cm.sup.3 to 1×10.sup.16

atoms/cm.sup.3, for example, 1×10.sup.14 atoms/cm.sup.3, 5×10.sup.14 atoms/cm.sup.3, 1×10.sup.15 atoms/cm.sup.3, 5×10.sup.15 atoms/cm.sup.3, or 1×10.sup.16 atoms/cm.sup.3. In some embodiments, an ion implantation energy used in the third ion implantation process **209** is larger than the ion implantation energy used in the second ion implantation process **206**. In some embodiments, the ion implantation energy used in the third ion implantation process **209** is preferably from 10 KeV to 15 KeV, for example, 10 KeV, 11 KeV, 12 KeV, 13 KeV, 14 KeV, or 15 KeV, such that a desirable depth (e.g., the depths D**5**′ described above) of the third implantation regions can be achieved. In some embodiments, after performing the third ion implantation process **209**, the first electrodes **105** is formed on the third implantation regions to obtain the semiconductor device **10** shown in FIG. **1**.

[0050] In some embodiments, when a plurality of units **100** described above is formed, the first implantation region **202** formed in the operation **21** extends between different units **100** as a region indicated by the dashed line **108** shown in FIG. **2**. In addition, the material **203** of the gate structure **102**G and the gate structure **103**G formed in the operation **22** extends along the first direction X and the second direction Y, and after patterning the material **203** by the patterned photoresist layer **204**, the remained material **203** is shown in FIG. **2**, in which the gate structures **102**G of the electronic fuses **102** in different units **100** are physically connected, and the gate structures **103**G of the switching transistors **103** in different units **100** are physically connected. In addition, the second electrodes **106** are formed on portions of the remained material **203** physically connected to the gate structures **102**G, and the third electrodes **107** are formed on portions of the remained material **203** physically connected to the gate structures **103**G.

[0051] The semiconductor device and the method of forming the same avoid the diffusion of the dopants to contaminate the doping regions respectively underneath the electronic fuse and the switching transistor. In addition, the size of the semiconductor device can be smaller without having the dopants contaminate the doping regions underneath the electronic fuse and the switching transistor. In addition, the method is much easier and does not exceed the limitations of manufacturing tools nowadays.

[0052] The present disclosure is described in considerable detail in some embodiments, but other embodiments may also be feasible, so the description of the embodiments in the present disclosure is not intended to limit the scope and spirit of the claims attached. For one skilled in the art, the present disclosure may be modified and changed without deviating from the scope and spirit of the present disclosure. Such modifications and changes are intended to be covered by the present disclosure when they belong to the scope and spirit of the attached claims.

Claims

1. A semiconductor device, comprising: a substrate; an electronic fuse on the substrate; and a switching transistor on the substrate and next to the electronic fuse, wherein: a first doping region of the substrate underneath the electronic fuse has a first conductive type, a second doping region of the substrate underneath the switching transistor has a second conductive type, and the first conductive type and the second conductive type are the same, and; the first doping region of the substrate is disposed between third doping regions of the substrate, and third conductive types of the third doping regions are different from the first conductive type.

2. The semiconductor device of claim 1, wherein the first doping region and the third doping regions are disposed between source/drain doping regions of the substrate.

3. The semiconductor device of claim 2, wherein depths of the source/drain doping regions in the substrate are larger than depths of the third doping regions in the substrate.

4. The semiconductor device of claim 2, wherein dopant concentrations of the source/drain doping regions are larger than dopant concentrations of the third doping regions.

5. The semiconductor device of claim 2, wherein fourth conductive types of the source/drain

doping regions are the same as the third conductive types.
6. The semiconductor device of claim 2, wherein the second doping region of the substrate is disposed between fourth doping regions of the substrate, fourth conductive types of the fourth doping regions are different from the second conductive type, and one of the source/drain doping regions extends continuously from one of the third doping regions to one of the fourth doping regions.
7. The semiconductor device of claim 1, wherein the first doping region is in direct contact with the third doping regions.
8. The semiconductor device of claim 1, wherein the electronic fuse comprises a gate structure, first spacers on sidewalls of the gate structure, and second spacers on the first spacers.
9. The semiconductor device of claim 8, wherein the first doping region is disposed below the gate structure and the first spacers, and the third doping regions are disposed below the second spacers.
10. The semiconductor device of claim 1, wherein: the first conductive type is a P type, the second conductive type is a P type, and the third conductive types are N types; or the first conductive type is an N type, the second conductive type is an N type, and the third conductive types are P types.
11. The semiconductor device of claim 1, wherein a minimum distance between a gate structure of the electronic fuse and a gate structure of the switching transistor is smaller than 100 nm.
12. A method of forming semiconductor device, comprising: performing a first ion implantation process to form a first implantation region in a substrate; forming a gate structure of an electronic fuse and a gate structure of a switching transistor on the first implantation region; forming first spacers on sidewalls of the gate structure of the electronic fuse; performing a second ion implantation process adjacent to the first spacers to transform first portions of the first implantation region into second implantation regions in the substrate, wherein conductive types of the second implantation regions are different from a conductive type of the first implantation region; forming second spacers on the first spacers; and performing a third ion implantation process adjacent to the second spacers to transform first portions of the second implantation regions into third implantation regions in the substrate, wherein conductive types of the third implantation regions are the same as conductive types of the second implantation regions.
13. The method of claim 12, wherein after performing the third ion implantation process, a second portion of the first implantation region is remained below the gate structure and the first spacers, second portions of the second implantation regions are remained below the second spacers, and the second portion of the first implantation region and the second portions of the second implantation regions are located between the third implantation regions.
14. The method of claim 12, wherein an ion implantation energy used in the third ion implantation process is larger than an ion implantation energy used in the second ion implantation process.
15. The method of claim 12, wherein a dopant concentration used in the second ion implantation process is from 1×10.sup.13 atoms/cm.sup.3 to 1×10.sup.15 atoms/cm.sup.3, and a dopant concentration used in the third ion implantation process is from 1×10.sup.14 atoms/cm.sup.3 to 1×10.sup.16 atoms/cm.sup.3.
16. The method of claim 12, wherein: forming the first spacers further comprises forming the first spacers on sidewalls of the gate structure of the switching transistor; and after performing the third ion implantation process, second portions of the second implantation regions are remained and located adjacent to the gate structure of the electronic fuse, third portions of the second implantation regions are remained and located adjacent to the gate structure of the switching transistor, and one of the third implantation regions extends continuously from one of the second portions of the second implantation regions to one of the third portions of the second implantation regions.
17. The method of claim 12, wherein: a P type dopant is used in the first ion implantation process, and N type dopants are used in the second ion implantation process and the third ion implantation process; or an N type dopant is used in the first ion implantation process, and P type dopants are

used in the second ion implantation process and the third ion implantation process.
18. The method of claim 12, wherein a dopant concentration used in the first ion implantation process is from 1×10.sup.11 atoms/cm.sup.3 to 1×10.sup.14 atoms/cm.sup.3.